Application deployment (Deployment of React application to a product ready state)

Project Overview

This document outlines the complete deployment process of a React application from development to a production-ready environment using Git, Docker, Jenkins, AWS EC2, and monitoring tools.

Important Links :

🔗 GitHub repo → https://github.com/abinayanachimuthu123-beep/devops-build

🌐 Deployed site → http://34.219.209.243

🐳 Docker HUB: [Repositories | abinayacould]

🐳 Dev image → abinayacould/dev:latest

🐳 Prod image → abinayacould/prod_repo:latest

📊 Monitoring → http://34.219.209.243:3001

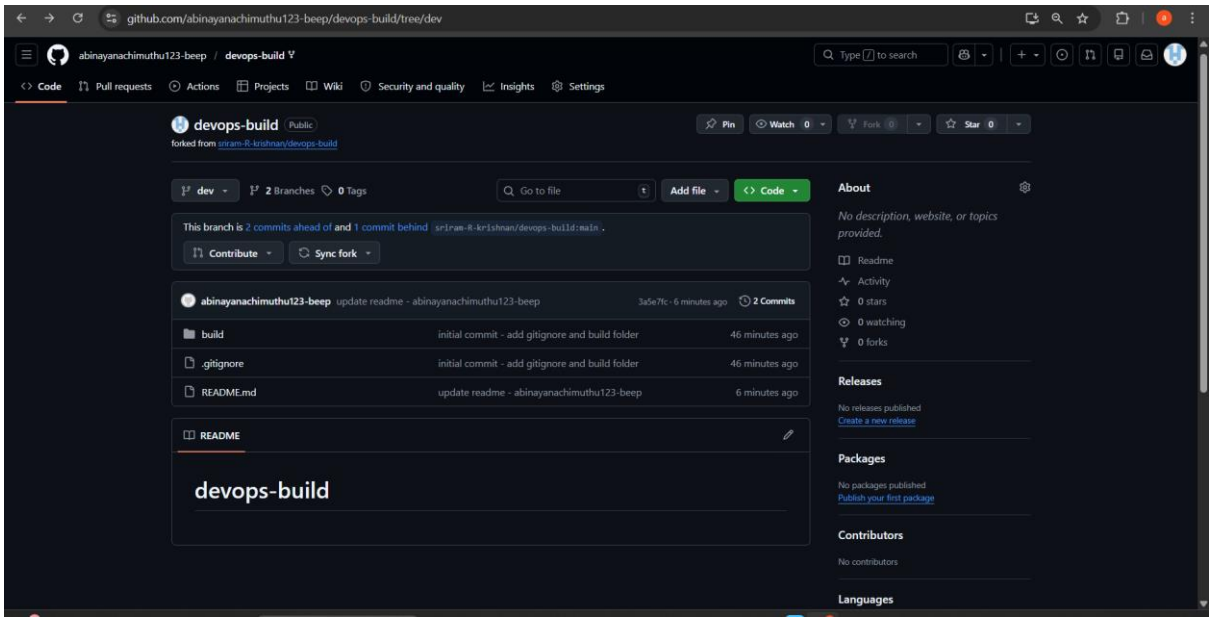
Git Cloning & Source Code Management (Git & GitHub) :

1. Installed Git and used **Git Bash** for executing commands.
2. Cloned the repository to the local system for development:

```
git clone https://github.com/abinayanachimuthu123-beep/devops-build.git
cd devops-build
```

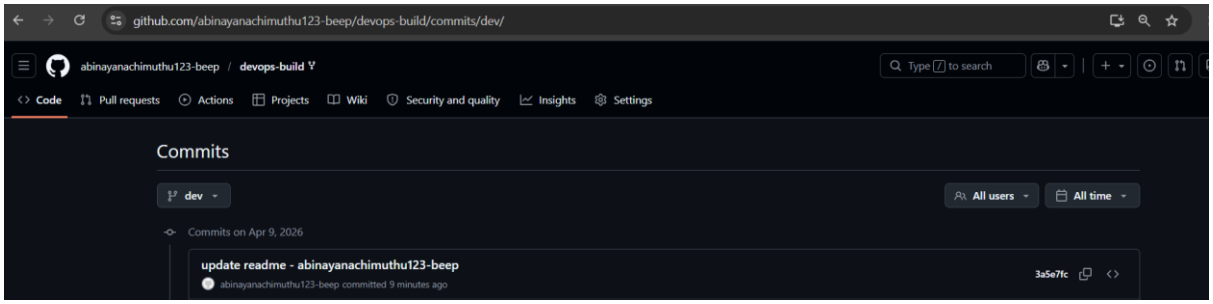
3. Followed a structured **branching strategy**:
   - dev → Development
   - main → Production
4. Created a .gitignore file to exclude sensitive and unnecessary files such as:
   - .env

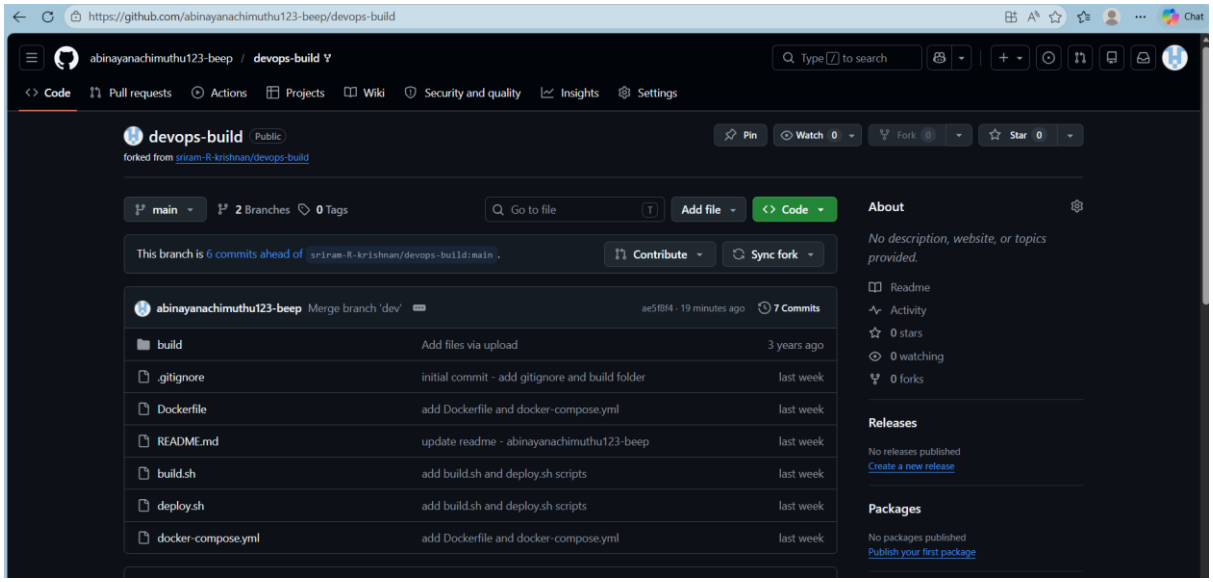
- node_modules/




```
 create mode 100644 docker-compose.yml

abina@LAPTOP-NAK4ILBA MINGW64 ~/Projects/devops-build (main)
$ git push origin main
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 267 bytes | 267.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/abinayanachimuthu123-beep/devops-build.git
   ef44730..ae5f8f4  main -> main

abina@LAPTOP-NAK4ILBA MINGW64 ~/Projects/devops-build (main)
$ |
```

```
abina@LAPTOP-NAK4ILBA MINGW64 ~/Projects
$ git clone https://github.com/abinayanachimuthu123-beep/devops-build.git
Cloning into 'devops-build'...
remote: Enumerating objects: 38, done.
remote: Counting objects: 100% (15/15), done.
remote: Compressing objects: 100% (12/12), done.
remote: Total 38 (delta 3), reused 13 (delta 1), pack-reused 23 (from 1)
Receiving objects: 100% (38/38), 721.89 KiB | 3.15 MiB/s, done.
Resolving deltas: 100% (3/3), done.

abina@LAPTOP-NAK4ILBA MINGW64 ~/Projects
$ ls -la
total 12
drwxr-xr-x 1 abina 197609 0 Apr 16 10:09 ./
drwxr-xr-x 1 abina 197609 0 Apr 16 10:08 ../
drwxr-xr-x 1 abina 197609 0 Apr 16 10:09 devops-build/
```





Containerization Using Docker :

1 - Verified Docker installation.
2 -Created Dockerfile using Nginx.
3- Created docker-compose.yml to run on port 80.
4- Built image and started container.
5- Application accessible in browser.

```
MINGW64:/c/Users/abina/projects/devops-build-main/devops-build-main

abina@LAPTOP-NAK4ILBA MINGW64 ~/projects/devops-build-main/devops-build-main (dev)
$ docker --version
Docker version 29.2.1, build a5c7197

abina@LAPTOP-NAK4ILBA MINGW64 ~/projects/devops-build-main/devops-build-main (dev)
$ docker ps
failed to connect to the docker API at npipe:////./pipe/dockerDesktopLinuxEngine; check if the path is correct
le specified.

abina@LAPTOP-NAK4ILBA MINGW64 ~/projects/devops-build-main/devops-build-main (dev)
$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES

abina@LAPTOP-NAK4ILBA MINGW64 ~/projects/devops-build-main/devops-build-main (dev)
$ pwd
/c/Users/abina/projects/devops-build-main/devops-build-main

abina@LAPTOP-NAK4ILBA MINGW64 ~/projects/devops-build-main/devops-build-main (dev)
$ notepad Dockerfile
FROM nginx:alpine
COPY build/ /usr/share/nginx/html
EXPOSE 80
CMD ["nginx", "-g", "daemon off;"]


abina@LAPTOP-NAK4ILBA MINGW64 ~/projects/devops-build-main/devops-build-main (dev)
$ cat Dockerfile
cat: Dockerfile: No such file or directory

abina@LAPTOP-NAK4ILBA MINGW64 ~/projects/devops-build-main/devops-build-main (dev)
$ notepad Dockerfile

abina@LAPTOP-NAK4ILBA MINGW64 ~/projects/devops-build-main/devops-build-main (dev)
$ pwd
/c/Users/abina/projects/devops-build-main/devops-build-main

abina@LAPTOP-NAK4ILBA MINGW64 ~/projects/devops-build-main/devops-build-main (dev)
$ cat Dockerfile
cat: Dockerfile: No such file or directory

abina@LAPTOP-NAK4ILBA MINGW64 ~/projects/devops-build-main/devops-build-main (dev)
$ ^C

abina@LAPTOP-NAK4ILBA MINGW64 ~/projects/devops-build-main/devops-build-main (dev)
$ cat > Dockerfile << 'EOF'
FROM nginx:alpine
COPY build/ /usr/share/nginx/html
EXPOSE 80
CMD ["nginx", "-g", "daemon off;"]
EOF

abina@LAPTOP-NAK4ILBA MINGW64 ~/projects/devops-build-main/devops-build-main (dev)
$ cat Dockerfile
FROM nginx:alpine
COPY build/ /usr/share/nginx/html
EXPOSE 80
CMD ["nginx", "-g", "daemon off;"]
```

```
abina@LAPTOP-NAK4ILBA MINGW64 ~/projects/devops-build-main/devops-build-main (dev)
$ cat Dockerfile
FROM nginx:alpine
COPY build/ /usr/share/nginx/html
EXPOSE 80
CMD ["nginx", "-g", "daemon off;"]

abina@LAPTOP-NAK4ILBA MINGW64 ~/projects/devops-build-main/devops-build-main (dev)
$ notepad docker-compose.yml

abina@LAPTOP-NAK4ILBA MINGW64 ~/projects/devops-build-main/devops-build-main (dev)
$ cat docker-compose.yml
version: '3'
services:
  web:
    build: .
    ports:
      - "80:80"
abina@LAPTOP-NAK4ILBA MINGW64 ~/projects/devops-build-main/devops-build-main (dev)
```

Docker Image:

1.Building an image from Dockerfile

```
      - "80:80"
abina@LAPTOP-NAK4ILBA MINGW64 ~/projects/devops-build-main/devops-build-main (dev)
$ docker build -t devops-build .
[+] Building 10.9s (8/8) FINISHED
 => [internal] load build definition from Dockerfile
 => => transferring dockerfile: 134B
 => [internal] load metadata for docker.io/library/nginx:alpine
 => [auth] library/nginx:pull token for registry-1.docker.io
 => [internal] load .dockerignore
 => => transferring context: 2B
 => [internal] load build context
 => => transferring context: 2.61MB
 => [1/2] FROM docker.io/library/nginx:alpine@sha256:645eda1c2477aaa9b879f73909b9222c6f19798dd45be6706268d82a661c6e6d
 => => resolve docker.io/library/nginx:alpine@sha256:645eda1c2477aaa9b879f73909b9222c6f19798dd45be6706268d82a661c6e6d
 => => sha256:1165b869c51a1a0747d78cec8fab96c30156a979e51ecf2f91aa792e557d94a4 20.25MB / 20.25MB
 => => sha256:34dfdd2ef1f920d0054dde2fc09ddc83ff8e71d05fadb79e2cab6e6234596f0a 1.21kB / 1.21kB
 => => sha256:c8a2fa3a88d244a3f32dcbc9c1f7649c662661a28c624198ada43aa0b7598e7f 1.40kB / 1.40kB
 => => sha256:a71873b303e8d75170b7ced2725b01b3ae15ad76f0d4eef16a49335821b6a0ef 404B / 404B
 => => sha256:ff9f59a6a62e9e9f29d7a84fb18865b45664d3f0d061eff7548bd61746dd101c 957B / 957B
 => => sha256:15e759724ff67f262e38bb7c070af9d0b84f959f9b37fa966f68bf2f881a4b62 627B / 627B
 => => sha256:f03becc8ac15611cfcc421c977a5ba4d65456093570788523a4ba557689aa7f7 1.87MB / 1.87MB
 => => sha256:589002ba0eaed121a1dbf42f6648f29e5be55d5c8a6ee0f8eaa0285cc21ac153 3.86MB / 3.86MB
 => => extracting sha256:589002ba0eaed121a1dbf42f6648f29e5be55d5c8a6ee0f8eaa0285cc21ac153
 => => extracting sha256:f03becc8ac15611cfcc421c977a5ba4d65456093570788523a4ba557689aa7f7
 => => extracting sha256:15e759724ff67f262e38bb7c070af9d0b84f959f9b37fa966f68bf2f881a4b62
 => => extracting sha256:ff9f59a6a62e9e9f29d7a84fb18865b45664d3f0d061eff7548bd61746dd101c
 => => extracting sha256:a71873b303e8d75170b7ced2725b01b3ae15ad76f0d4eef16a49335821b6a0ef
 => => extracting sha256:34dfdd2ef1f920d0054dde2fc09ddc83ff8e71d05fadb79e2cab6e6234596f0a
 => => extracting sha256:c8a2fa3a88d244a3f32dcbc9c1f7649c662661a28c624198ada43aa0b7598e7f
 => => extracting sha256:1165b869c51a1a0747d78cec8fab96c30156a979e51ecf2f91aa792e557d94a4
 => [2/2] COPY build/ /usr/share/nginx/html
 => exporting to image
 => => exporting layers
 => => exporting manifest sha256:6948f12bdb802531f21cf71135c367da4a8b125616ee34dfc5952fca8ddaedbc
 => => exporting config sha256:f73637477ca1b1c6044acb39cb16efb704b14fcd725731719b3d0691240c27d8
 => => exporting attestation manifest sha256:41291706f71867ef738ee7d4bb50971d219aeb1f3d6bb033fd9cacbcd724a874
 => => exporting manifest list sha256:13008245d0a36d836f13cfb6dca0b9ac6d7e7e99ea398a6339109cd550914f26
 => => naming to docker.io/library/devops-build:latest
 => => unpacking to docker.io/library/devops-build:latest

abina@LAPTOP-NAK4ILBA MINGW64 ~/projects/devops-build-main/devops-build-main (dev)
$
```

```
abina@LAPTOP-NAK4ILBA MINGW64 ~/projects/devops-build-main/devops-build-main (dev)
$ docker images

IMAGE                                                                     ID             DISK USAGE   CONTENT SIZE   EXTRA
devops-build:latest                                                       13008245d0a3         96MB         26.7MB
gcr.io/k8s-minikube/kicbase:v0.0.49                                       c7ddf4818024       1.91GB          539MB
gcr.io/k8s-minikube/kicbase:v0.0.50                                       b97074569ae9       1.94GB          545MB
gcr.io/k8s-minikube/kicbase@sha256:e6daddbb1dc09ccd195c5605f65e2d38406c36ef36c5a492ffe805d9d36f4945  e6daddbb1dc0       1.91GB          539MB
gcr.io/k8s-minikube/kicbase@sha256:eb4fec00e8ad70adf8e6436f195cc429825ffb85f95afcdb5d8d9deb576f3e93  eb4fec00e8ad       1.94GB          545MB
hello-world:latest                                                        85404b3c5395       25.9kB         9.52kB    U

abina@LAPTOP-NAK4ILBA MINGW64 ~/projects/devops-build-main/devops-build-main (dev)
$
```

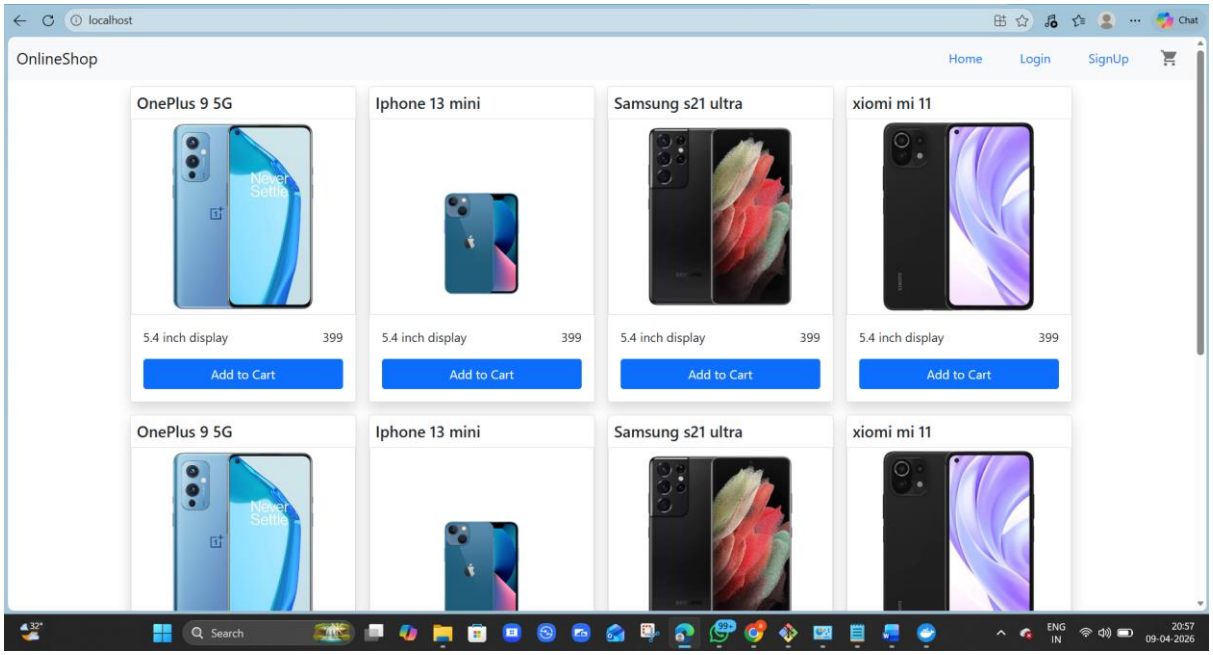
2.Image devops-build-main-web — **Built**

3.Network devops-build-main_default — **Created**

4.Container devops-build-main-web-1 — **Started**

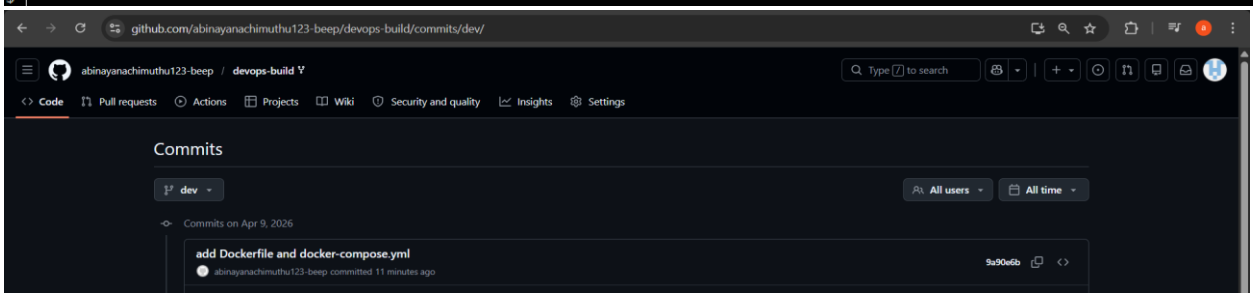
```
abina@LAPTOP-NAK4ILBA MINGW64 ~/projects/devops-build-main/devops-build-main (dev)
$ docker-compose up -d
time="2026-04-09T12:28:10+05:30" level=warning msg="C:\\Users\\abina\\projects\\devops-build-main\\devops-build-main\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove i
t to avoid potential confusion"
#1 [internal] load local bake definitions
#1 reading from stdin 589B done
#1 DONE 0.0s

#2 [internal] load build definition from Dockerfile
#2 transferring dockerfile: 134B 0.1s done
#2 DONE 0.1s

#3 [internal] load metadata for docker.io/library/nginx:alpine
#3 ...

#4 [auth] library/nginx:pull token for registry-1.docker.io
#4 DONE 0.0s

#3 [internal] load metadata for docker.io/library/nginx:alpine
#3 DONE 2.0s

#5 [internal] load .dockerignore
#5 transferring context: 2B done
#5 DONE 0.0s

#6 [internal] load build context
#6 transferring context: 837B 0.0s done
#6 DONE 0.0s

#7 [1/2] FROM docker.io/library/nginx:alpine@sha256:645eda1c2477aaa9b879f73909b9222c6f19798dd45be6706268d82a661c6e6d
#7 resolve docker.io/library/nginx:alpine@sha256:645eda1c2477aaa9b879f73909b9222c6f19798dd45be6706268d82a661c6e6d 0.0s done
#7 DONE 0.1s

#8 [2/2] COPY build/ /usr/share/nginx/html
#8 CACHED

#9 exporting to image
#9 exporting layers done
#9 exporting manifest sha256:e1a90458af3244c155419abf7c76d34276971ec2677a6ec908d7d86144c7feb9 0.0s done
#9 exporting config sha256:5d1750446f7ca84194e1cc82701500723a435c2de3629c9c10574edbb20517f4 0.0s done
#9 exporting attestation manifest sha256:3a794ca611091ddc55d1e0ca2fb6759ef63730546ecc29bf4666228a20ca3bde 0.0s done
#9 exporting manifest list sha256:26e6794754c787dfda304e137c3ff96c57a9949b31f18e23e9602c7a60cb3c06
#9 exporting manifest list sha256:26e6794754c787dfda304e137c3ff96c57a9949b31f18e23e9602c7a60cb3c06 0.0s done
#9 naming to docker.io/library/devops-build-main-web:latest done
#9 unpacking to docker.io/library/devops-build-main-web:latest 0.0s done
#9 DONE 0.2s

#10 resolving provenance for metadata file
#10 DONE 0.0s
[+] up 3/3
 ✔ Image devops-build-main-web       Built                                                                          3.7s
 ✔ Network devops-build-main_default Created                                                                        0.1s
 ✔ Container devops-build-main-web-1 Started                                                                        1.3s

abina@LAPTOP-NAK4ILBA MINGW64 ~/projects/devops-build-main/devops-build-main (dev)
$
```

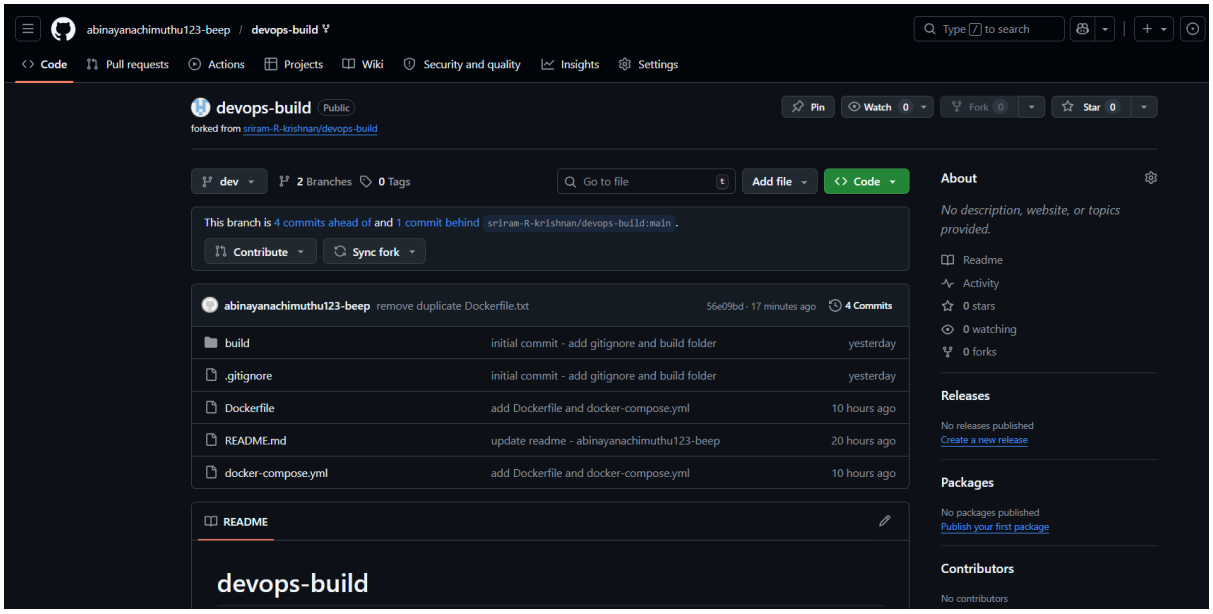
5.React app live in the browser



6.commited the Dockerfile and docker-compose.yml to GitHub :

```
abina@LAPTOP-NAK4ILBA MINGW64 ~/projects/devops-build-main/devops-build-main (dev)
$ git status
On branch dev
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        Dockerfile
        Dockerfile.txt
        docker-compose.yml

nothing added to commit but untracked files present (use "git add" to track)

abina@LAPTOP-NAK4ILBA MINGW64 ~/projects/devops-build-main/devops-build-main (dev)
$ git add .
warning: in the working copy of 'Dockerfile', LF will be replaced by CRLF the next time Git touches it

abina@LAPTOP-NAK4ILBA MINGW64 ~/projects/devops-build-main/devops-build-main (dev)
$ git commit -m "add Dockerfile and docker-compose.yml"
[dev 9a90e6b] add Dockerfile and docker-compose.yml
 3 files changed, 14 insertions(+)
 create mode 100644 Dockerfile
 create mode 100644 Dockerfile.txt
 create mode 100644 docker-compose.yml

abina@LAPTOP-NAK4ILBA MINGW64 ~/projects/devops-build-main/devops-build-main (dev)
$ git push origin dev
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 12 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 618 bytes | 618.00 KiB/s, done.
Total 5 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), done.
To https://github.com/abinayanachimuthu123-beep/devops-build.git
   3a5e7fc..9a90e6b  dev -> dev

abina@LAPTOP-NAK4ILBA MINGW64 ~/projects/devops-build-main/devops-build-main (dev)
$
```





Bash :

1.Created build.sh — automates Docker image building

2.Created deploy.sh — automates container deployment

3.Tested both scripts — both worked perfectly

4.Pushed to GitHub dev branch

```
abina@LAPTOP-NAK4ILBA MINGW64 ~/projects/devops-build-main/devops-build-main (dev)
$ notepad build.sh

abina@LAPTOP-NAK4ILBA MINGW64 ~/projects/devops-build-main/devops-build-main (dev)
$ cat build.sh
#!/bin/bash
echo "Building Docker image..."
docker build -t devops-build .
echo "Build completed successfully!"
abina@LAPTOP-NAK4ILBA MINGW64 ~/projects/devops-build-main/devops-build-main (dev)
$
```

```
abina@LAPTOP-NAK4ILBA MINGW64 ~/projects/devops-build-main/devops-build-main (dev)
$ notepad deploy.sh

abina@LAPTOP-NAK4ILBA MINGW64 ~/projects/devops-build-main/devops-build-main (dev)
$ cat deploy.sh
#!/bin/bash
echo "Deploying application..."
docker-compose down
docker-compose up -d
echo "Application deployed successfully!"
abina@LAPTOP-NAK4ILBA MINGW64 ~/projects/devops-build-main/devops-build-main (dev)
$
```

```
abina@LAPTOP-NAK4ILBA MINGW64 ~/projects/devops-build-main/devops-build-main (dev)
$ ./build.sh
Building Docker image...
[+] Building 2.8s (8/8) FINISHED                                          docker:desktop-linux
 => [internal] load build definition from Dockerfile                                        0.1s
 => => transferring dockerfile: 134B                                                        0.1s
 => [internal] load metadata for docker.io/library/nginx:alpine                             2.1s
 => [auth] library/nginx:pull token for registry-1.docker.io                                0.0s
 => [internal] load .dockerignore                                                           0.0s
 => => transferring context: 2B                                                             0.0s
 => [internal] load build context                                                           0.0s
 => => transferring context: 837B                                                           0.0s
 => [1/2] FROM docker.io/library/nginx:alpine@sha256:645eda1c2477aaa9b879f73909b9222c6f19798dd45be  0.1s
 => => resolve docker.io/library/nginx:alpine@sha256:645eda1c2477aaa9b879f73909b9222c6f19798dd45be  0.1s
 => CACHED [2/2] COPY build/ /usr/share/nginx/html                                          0.0s
 => exporting to image                                                                      0.2s
 => => exporting layers                                                                     0.0s
 => => exporting manifest sha256:6948f12bdb802531f21cf71135c367da4a8b125616ee34dfc5952fca8ddaedbc   0.0s
 => => exporting config sha256:f73637477ca1b1c6044acb39cb16efb704b14fcd725731719b3d0691240c27d8     0.0s
 => => exporting attestation manifest sha256:a540ee38d59cc9432cc211399bf3e4b7248186e2a1d0564f44084  0.0s
 => => exporting manifest list sha256:bb881e600e8bc4ad0469d8529bc50d2d2977449c12ec027699733147f2c3  0.0s
 => => naming to docker.io/library/devops-build:latest                                      0.0s
 => => unpacking to docker.io/library/devops-build:latest                                   0.0s
Build completed successfully!
```

```
abina@LAPTOP-NAK4ILBA MINGW64 ~/projects/devops-build-main/devops-build-main (dev)
$ ./deploy.sh
Deploying application...
time="2026-04-10T08:04:55+05:30" level=warning msg="C:\\Users\\abina\\projects\\devops-build-main\\devops-build-main\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove i
t to avoid potential confusion"
[+] down 2/2
 ✔ Container devops-build-main-web-1 Removed                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 0.7s
 ✔ Network devops-build-main_default Removed                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 0.4s
time="2026-04-10T08:04:56+05:30" level=warning msg="C:\\Users\\abina\\projects\\devops-build-main\\devops-build-main\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove i
t to avoid potential confusion"
[+] up 2/2
 ✔ Network devops-build-main_default Created                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
time="2026-04-10T08:04:56+05:30" level=warning msg="C:\\Users\\abina\\projects\\devops-build-main\\devops-build-main\\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove i
t to avoid potential confusion"
[+] up 2/2
 ✔ Network devops-build-main_default Created                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 0.1s
 ✔ Container devops-build-main-web-1 Started                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                
Application deployed successfully!

abina@LAPTOP-NAK4ILBA MINGW64 ~/projects/devops-build-main/devops-build-main (dev)
$ |
```

```
abina@LAPTOP-NAK4ILBA MINGW64 ~/projects/devops-build-main/devops-build-main (dev)
$ git add .

abina@LAPTOP-NAK4ILBA MINGW64 ~/projects/devops-build-main/devops-build-main (dev)
$ git commit -m
error: switch `m' requires a value

abina@LAPTOP-NAK4ILBA MINGW64 ~/projects/devops-build-main/devops-build-main (dev)
$ git commit -m "add build.sh and deploy.sh scripts"
[dev d3bcb9b] add build.sh and deploy.sh scripts
 2 files changed, 9 insertions(+)
 create mode 100644 build.sh
 create mode 100644 deploy.sh

abina@LAPTOP-NAK4ILBA MINGW64 ~/projects/devops-build-main/devops-build-main (dev)
$ |
```

Showing 2 changed files with **9 additions** and **0 deletions**.

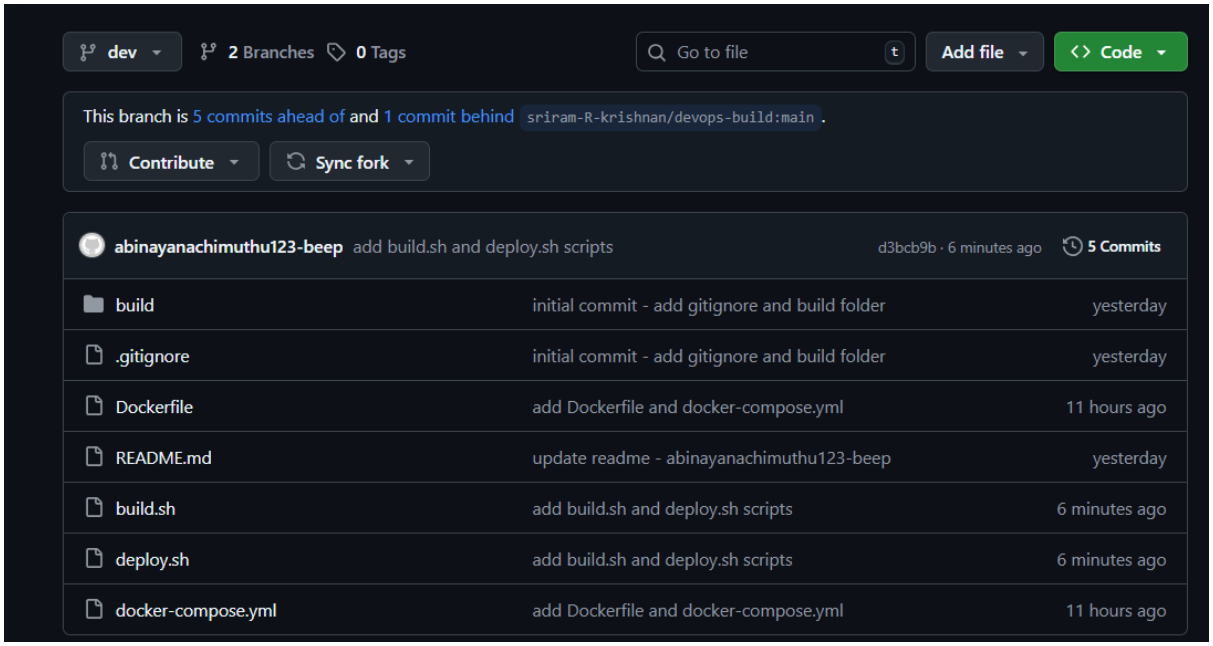
Split Unified

4 build.sh

```
@@ -0,0 +1,4 @@
+ #!/bin/bash
+ echo "Building Docker image..."
+ docker build -t devops-build .
+ echo "Build completed successfully!"
```

5 deploy.sh

```
@@ -0,0 +1,5 @@
+ #!/bin/bash
+ echo "Deploying application..."
+ docker-compose down
+ docker-compose up -d
+ echo "Application deployed successfully!"
```



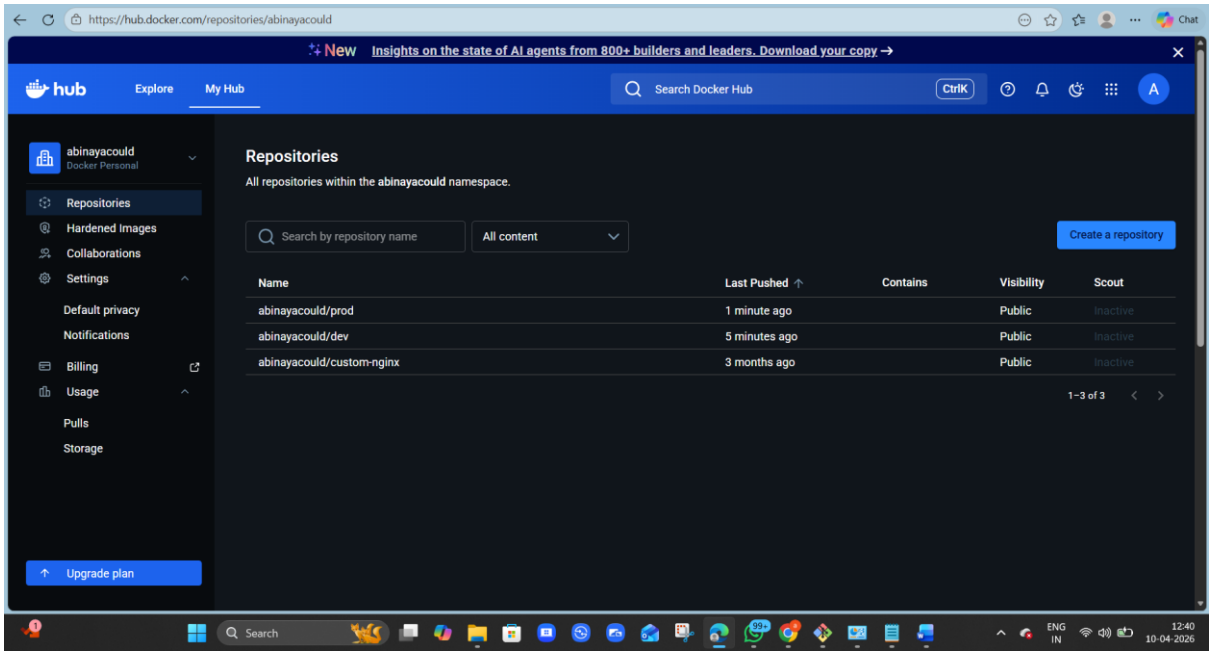
Docker Hub :

1.Created dev repo — Public on Docker Hub

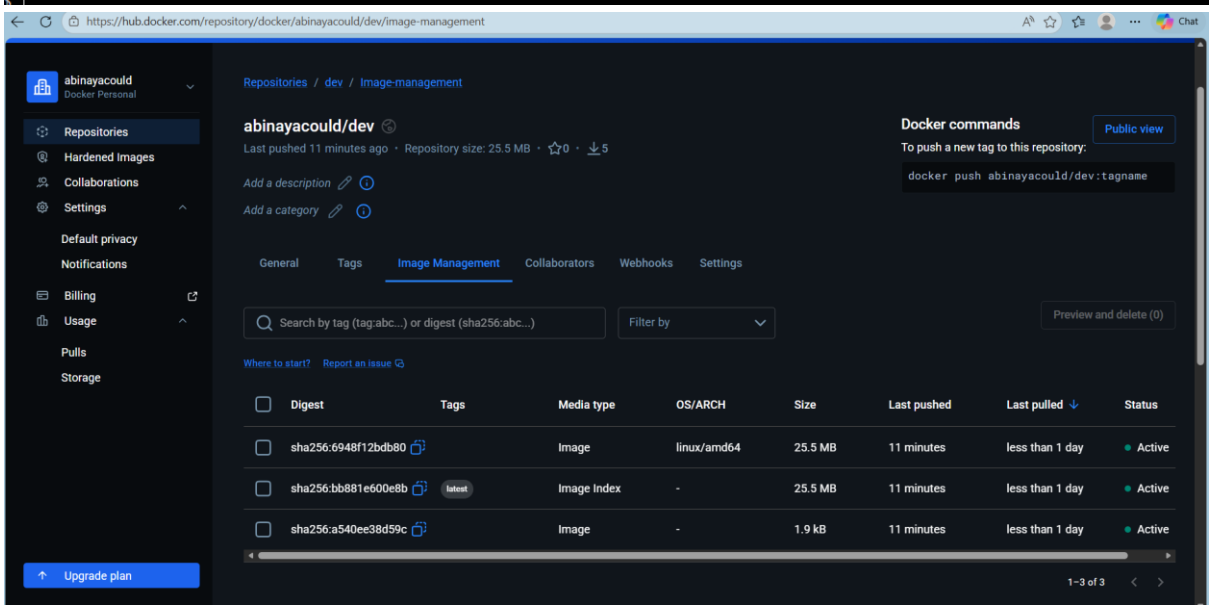
2.Created prod_repo repo — Private on Docker Hub

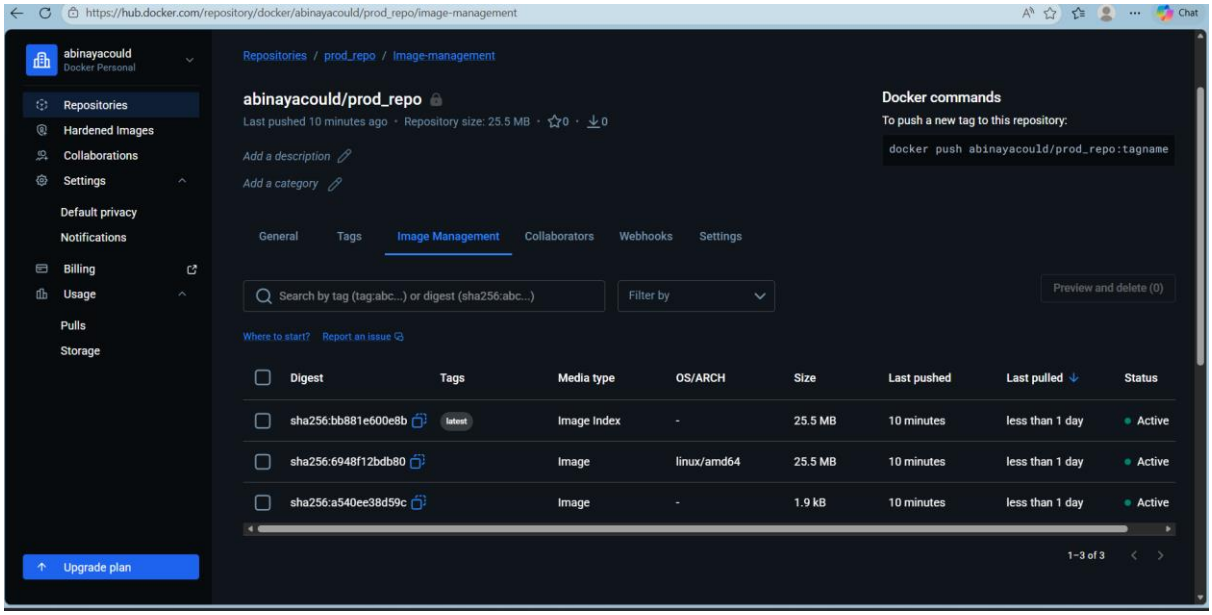
3.Pushed image to dev repo

4.Pushed image to prod repo



```
Login Succeeded

abina@LAPTOP-NAK4ILBA MINGW64 ~/projects/devops-build-main/devops-build-main (dev)
$ docker login
Authenticating with existing credentials... [Username: abinayacould]

i Info → To login with a different account, run 'docker logout' followed by 'docker login'


Login Succeeded

abina@LAPTOP-NAK4ILBA MINGW64 ~/projects/devops-build-main/devops-build-main (dev)
$ docker tag devops-build:latest abinayacould/dev:latest

abina@LAPTOP-NAK4ILBA MINGW64 ~/projects/devops-build-main/devops-build-main (dev)
$ docker push abinayacould/dev:latest
The push refers to repository [docker.io/abinayacould/dev]
589002ba0eae: Pushed
b957c941bb5b: Pushed
34dfdd2ef1f9: Pushed
c8a2fa3a88d2: Pushed
a71873b303e8: Pushed
6fd1b82dc0f4: Pushed
15e759724ff6: Pushed
ff9f59a6a62e: Pushed
1165b869c51a: Pushed
f03becc8ac15: Pushed
latest: digest: sha256:bb881e600e8bc4ad0469d8529bc50d2d2977449c12ec027699733147f2c3b51d size: 856

abina@LAPTOP-NAK4ILBA MINGW64 ~/projects/devops-build-main/devops-build-main (dev)
$ docker tag devops-build:latest abinayacould/prod_repo:latest

abina@LAPTOP-NAK4ILBA MINGW64 ~/projects/devops-build-main/devops-build-main (dev)
$ docker push abinayacould/prod_repo:latest
The push refers to repository [docker.io/abinayacould/prod_repo]
6fd1b82dc0f4: Mounted from abinayacould/dev
b957c941bb5b: Mounted from abinayacould/dev
ff9f59a6a62e: Mounted from abinayacould/dev
f03becc8ac15: Mounted from abinayacould/dev
34dfdd2ef1f9: Mounted from abinayacould/dev
c8a2fa3a88d2: Mounted from abinayacould/dev
1165b869c51a: Mounted from abinayacould/dev
589002ba0eae: Mounted from abinayacould/dev
15e759724ff6: Mounted from abinayacould/dev
a71873b303e8: Mounted from abinayacould/dev
latest: digest: sha256:bb881e600e8bc4ad0469d8529bc50d2d2977449c12ec027699733147f2c3b51d size: 856

abina@LAPTOP-NAK4ILBA MINGW64 ~/projects/devops-build-main/devops-build-main (dev)
$
```





Jenkins(CI\CD) :

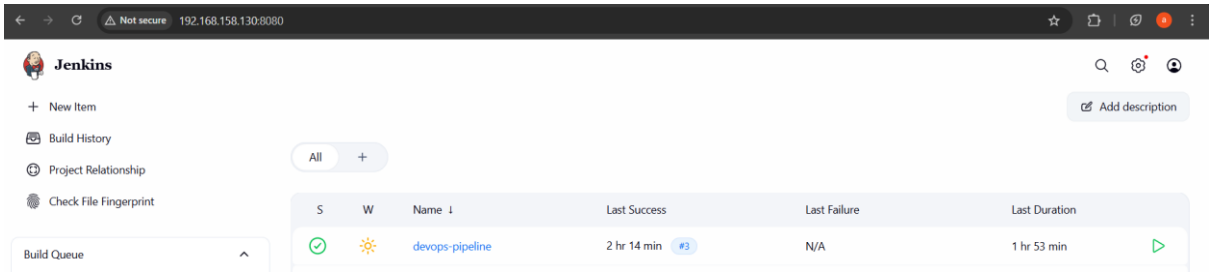
1.Jenkins pipeline created

2.Connected to GitHub repo

3.Auto builds Docker image

4.Auto pushes to Docker Hub dev repo

5.Pipeline runs successfully end to end



Pipeline

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script

Script ?

```
pipeline {
    agent any

    stages {
        stage('Clone') {
            steps {
                git branch: 'dev',
                    url: 'https://github.com/abinayanachimuthu123-beep/devops-build.git'
            }
        }

        stage('Build') {
            steps {
                sh 'chmod +x build.sh && ./build.sh'
            }
```

Use Groovy Sandbox ?

Pipeline

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script

Script ?

```
        }

        stage('Push to Dev Hub') {
            steps {
                sh '''
                    docker tag devops-build:latest abinayacould/dev:latest
                    docker push abinayacould/dev:latest
                '''
            }
        }

        stage('Deploy') {
            steps {
                sh 'chmod +x deploy.sh && ./deploy.sh'
            }
```

Use Groovy Sandbox ?

Pipeline

Define your Pipeline using Groovy directly or pull it from source control.

Definition

Pipeline script

Script ?

```
            steps {
                sh 'chmod +x deploy.sh && ./deploy.sh'
            }
        }
    }

    post {
        success {
            echo 'Pipeline completed successfully!'
        }
        failure {
            echo 'Pipeline failed!'
        }
    }
}
```

Use Groovy Sandbox ?

Pipeline Syntax

Not secure 192.168.158.130:8080/job/devops-pipeline/lastBuild/

Jenkins / devops-pipeline / #3

- Status
- Changes
- Console Output
- Edit Build Information
- Delete build '#3'
- Timings
- Git Build Data
- Pipeline Overview
- Restart from Stage
- Replay
- Pipeline Steps
- Workspaces

#3 (Apr 11, 2026, 2:51:06 AM)

Add description | Keep this build forever

Started by user Abinaya k N

Started 3 hr 18 min ago
Took 1 hr 53 min

This run spent:

- 1 hr 52 min waiting;
- 1 hr 53 min build duration;
- 1 hr 53 min total from scheduled to completion.

git
Revision: d3bcb9b34304f28d078d9d9f8b2dcf61e6d6a748
Repository: https://github.com/abinayanachimuthu123-beep/devops-build.git

- refs/remotes/origin/dev

No changes.

Jenkins / devops-pipeline / #3 / Stages

#3

Rerun

Started by Abinaya k N | Started 3 hr 22 min ago | Queued 0.1 sec | Took 1 hr 53 min

Graph

Start → Clone → Build → Push to Dev Hub → Deploy → Post Actions → End

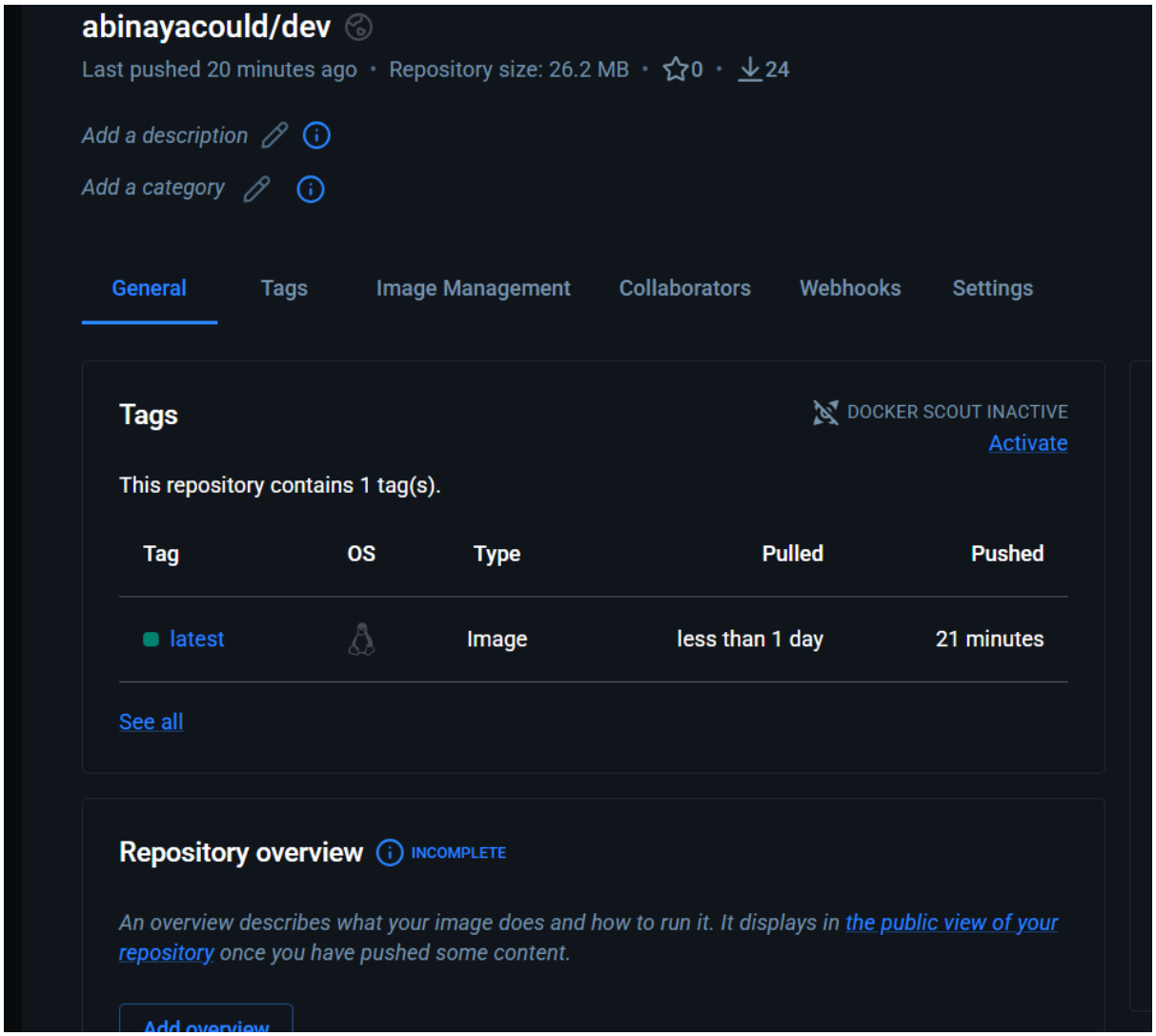
Search

- Clone 5s
- Build 23s
- Push to Dev Hub 9s
- Deploy 0.42s
- Post Actions 0.88s

Post Actions — 0.88s | Started 1h 30m ago | Jenkins

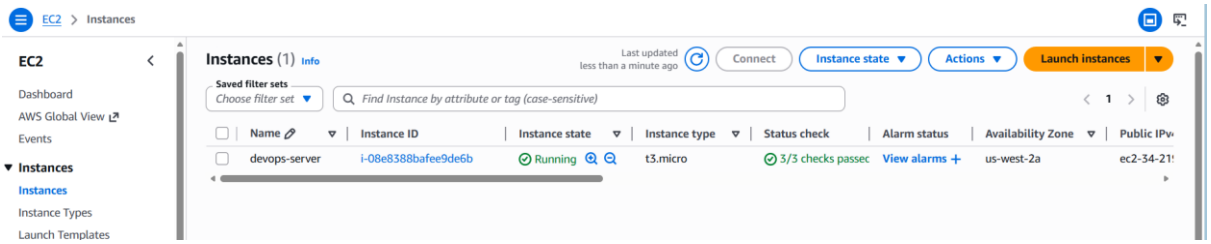
Pipeline completed successfully! — 0.29s

```
Pipeline completed successfully!
```

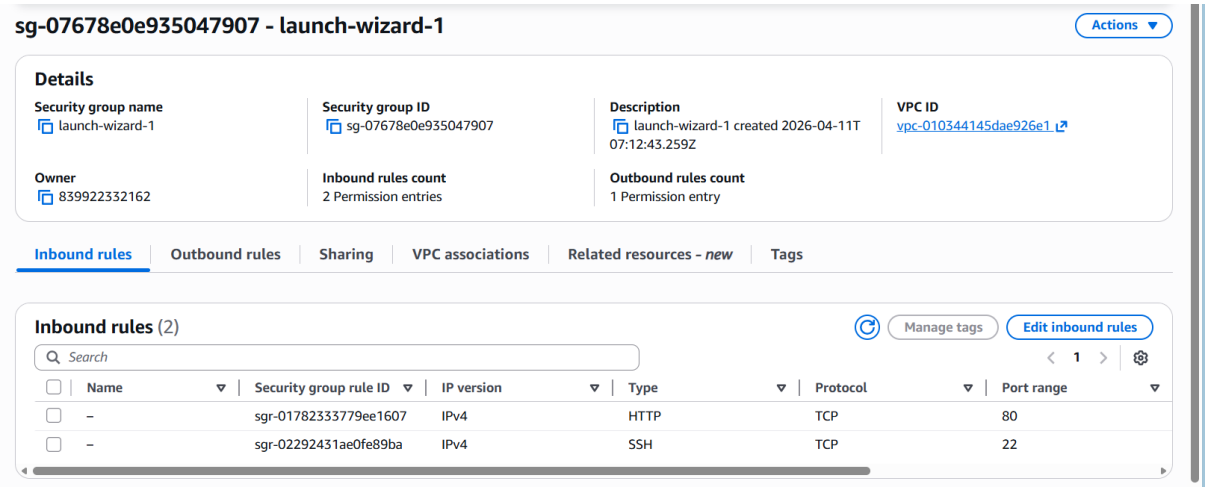


Deployment on AWS EC2 :

1.EC2 Instance created



2.Security Group is configured



3.CONNECTED TO YOUR AWS EC2 SERVER

```
abina@LAPTOP-NAK4ILBA MINGW64 ~/projects
$ ssh -i ~/projects/abinaya_oregon.pem ubuntu@34.219.209.243
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1007-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Sat Apr 11 07:39:20 UTC 2026

  System load:  0.0               Temperature:           -273.1 C
  Usage of /:   26.3% of 6.71GB   Processes:             110
  Memory usage: 24%               Users logged in:       0
  Swap usage:   0%                IPv4 address for ens5: 172.31.37.138

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The list of available updates is more than a week old.
To check for new updates run: sudo apt update


The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-37-138:~$
```

4.Docker is installed on EC2

```
sudo systemctl enable docker
ubuntu@ip-172-31-37-138:~$ docker --version
Docker version 29.1.3, build 29.1.3-0ubuntu3~24.04.1
ubuntu@ip-172-31-37-138:~$
```

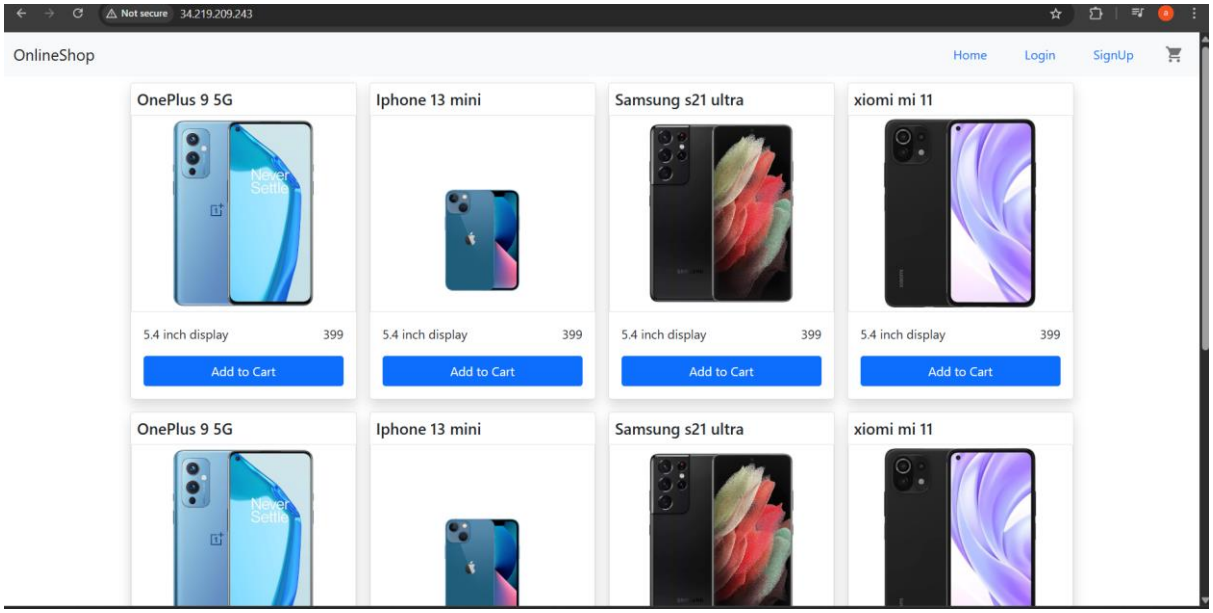
5.Image pulled successfully from Docker Hub to EC2

```
USING WEB-BASED LOGIN

i Info → To sign in with credentials on the command line, use 'docker login -u <username>'


Your one-time device confirmation code is: KKSL-SGMN
Press ENTER to open your browser or submit your device code here: https://login.docker.com/activate

Waiting for authentication in the browser…


WARNING! Your credentials are stored unencrypted in '/home/ubuntu/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
ubuntu@ip-172-31-37-138:~$ docker pull abinayacould/dev:latest
permission denied while trying to connect to the docker API at unix:///var/run/docker.sock
ubuntu@ip-172-31-37-138:~$ sudo usermod -aG docker ubuntu
ubuntu@ip-172-31-37-138:~$ newgrp docker
ubuntu@ip-172-31-37-138:~$ docker pull abinayacould/dev:latest
latest: Pulling from abinayacould/dev
f03becc8ac15: Pull complete
ff9f59a6a62e: Pull complete
a71873b303e8: Pull complete
589002ba0eae: Pull complete
f536610fc5f1: Pull complete
34dfdd2ef1f9: Pull complete
15e759724ff6: Pull complete
c8a2fa3a88d2: Pull complete
1165b869c51a: Pull complete
Digest: sha256:00c851a9f9e091cffdf2fca1475dc309e1097096e157de2b334463ac398ded2b
Status: Downloaded newer image for abinayacould/dev:latest
docker.io/abinayacould/dev:latest
ubuntu@ip-172-31-37-138:~$
```

6.CONTAINER IS RUNNING ON AWS EC2!

```
1165b869c51a: Pull complete
Digest: sha256:00c851a9f9e091cffdf2fca1475dc309e1097096e157de2b334463ac398ded2b
Status: Downloaded newer image for abinayacould/dev:latest
docker.io/abinayacould/dev:latest
ubuntu@ip-172-31-37-138:~$ docker run -d -p 80:80 abinayacould/dev:latest
681c3d017e88266514cf5bb7f5228d7ddf5e0777f24864575f23d9a170f42c48
ubuntu@ip-172-31-37-138:~$ docker ps
CONTAINER ID   IMAGE                  COMMAND                  CREATED          STATUS          PORTS                                   NAMES
681c3d017e88   abinayacould/dev:latest   "/docker-entrypoint.…"   21 seconds ago   Up 21 seconds   0.0.0.0:80->80/tcp, [::]:80->80/tcp   agitated_zhukovsky
ubuntu@ip-172-31-37-138:~$
```

7.APP IS LIVE ON THE INTERNET ON AWS EC2

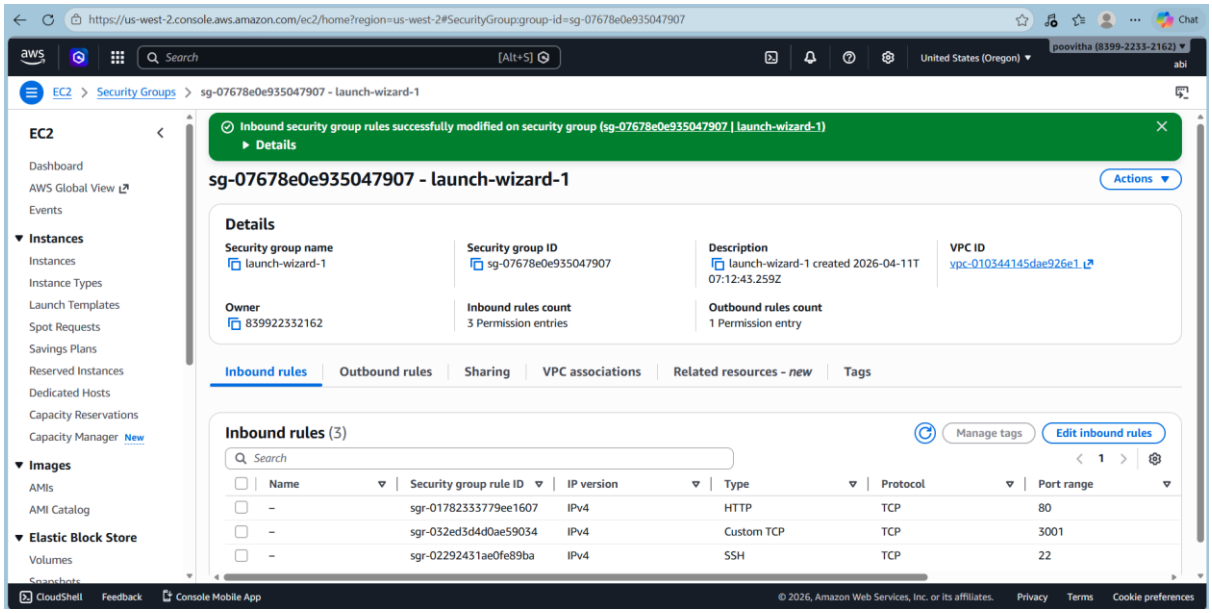


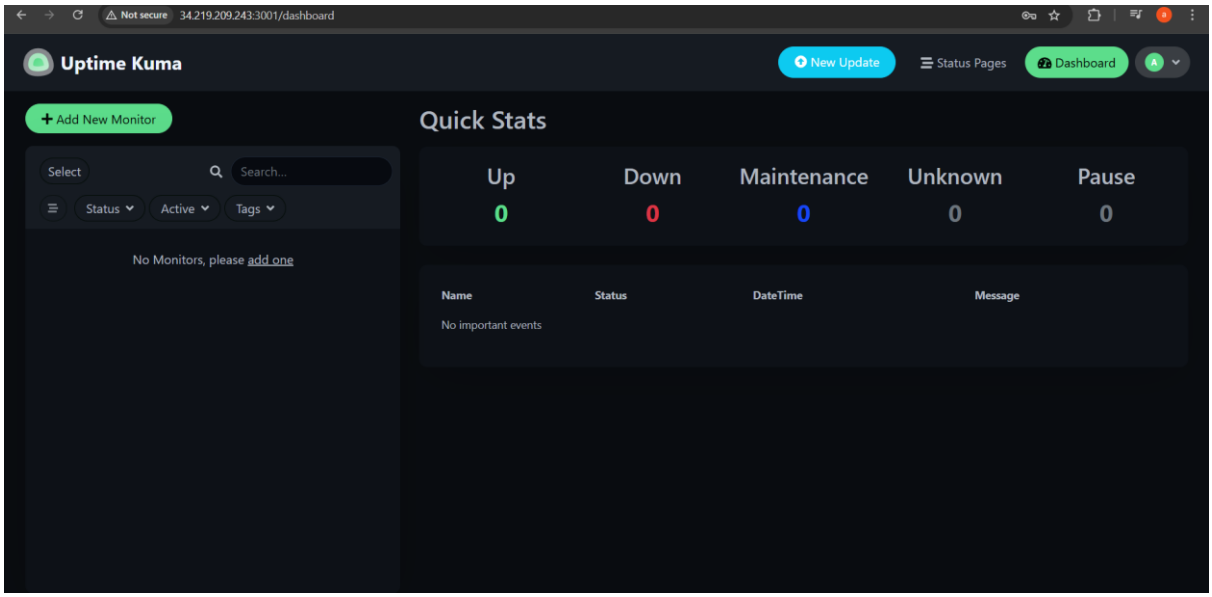
Monitoring Tool (Uptime Kuma)

1.Configured **Uptime Kuma** for monitoring

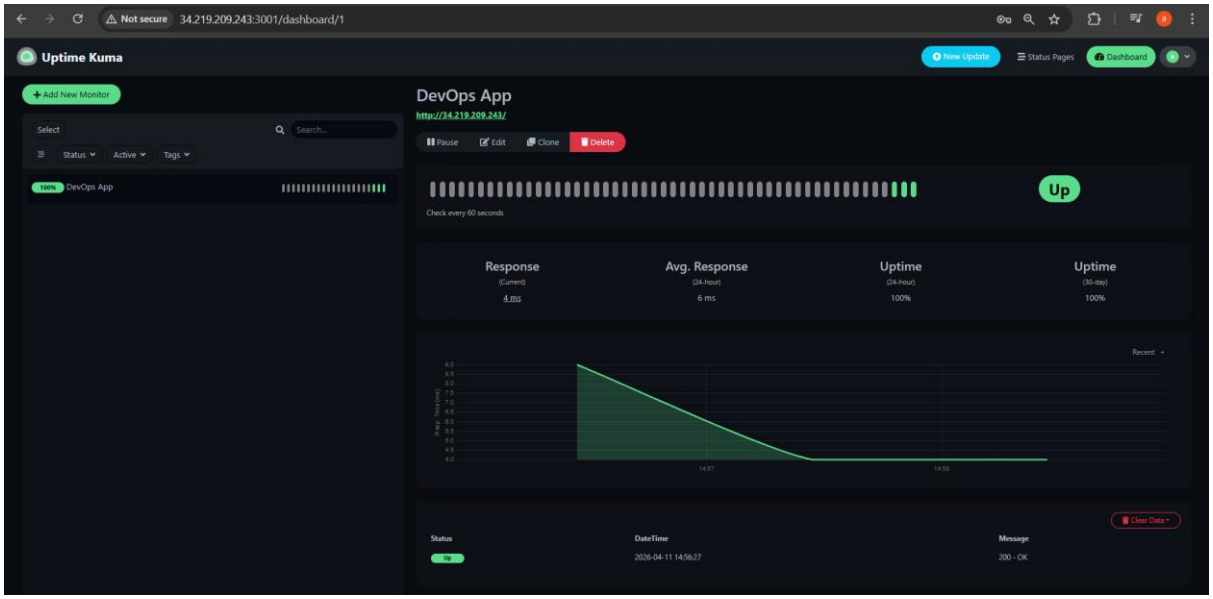
2.Health checks running every **60 seconds**

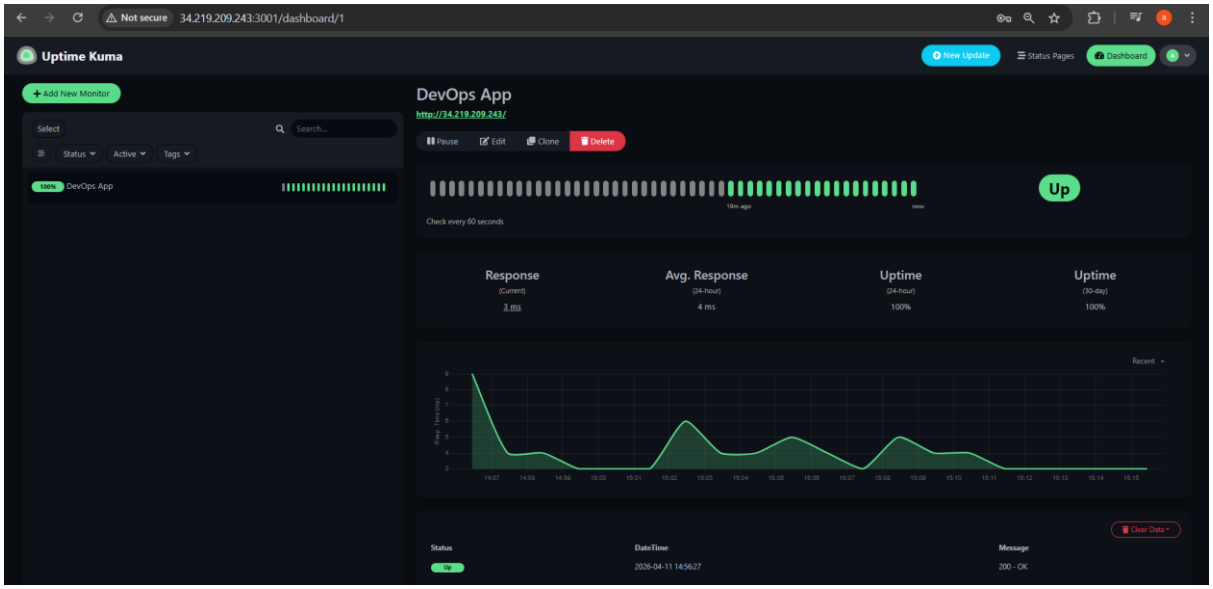
```
added 703 packages in 37s

141 packages are looking for funding
  run `npm fund` for details
ubuntu@ip-172-31-37-138:~$ uptime-kuma
Usage: uptime-kuma [options]

Options:
    --install-service    Install Uptime Kuma service (Windows and Linux only)
    --uninstall-service  Uninstall Uptime Kuma service
    --run                Run Uptime Kuma directly in the terminal
    --data-dir="your path"    Set the data directory
    --version            Print the version
    --help               Print this help
```





MONITORING IS LIVE(Checking every 60 seconds automatically!)





Final summary:

Part 1 — Git setup, forked & cloned repo, dev/master branches on GitHub

Part 2 — Dockerized the React app with Nginx

Part 3 — Automated build and deploy with Bash scripts

Part 4 — Docker Hub with dev (public) and prod (private) repos

Part 5 — Jenkins CI/CD pipeline auto-building on every push

Part 6 — App deployed live on AWS EC2 at http://34.219.209.243

Part 7 — Uptime Kuma monitoring with 100% uptime